Mihael Čudić, D.Phil.

Academic Curriculum Vitae

Instructor, Harvard Medical School · Investigator, Massachusetts General Hospital

Affiliation: Center for Precision Psychiatry, Massachusetts General Hospital
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RESEARCH PROFILE

Computational biomedical researcher developing and validating artificial intelligence, statistical genetics, and causal-inference methods for psychiatric research. Current work centers on longitudinal psychiatric phenotyping from clinical records, contextual interpretation of polygenic risk estimates, and integration of clinical and genomic information for psychiatric risk stratification.

ACADEMIC AND RESEARCH APPOINTMENTS

Current Appointments

Nov 2026–present **Instructor**, Harvard Medical School, Boston, MA, USA
Investigator, Center for Precision Psychiatry, Massachusetts General Hospital, Boston, MA, USA

Previous Academic Appointments

Jan–May 2024 **Adjunct Dunlevie Honors Professor**, University of Florida, Gainesville, FL, USA

RESEARCH AFFILIATIONS

Jan 2025–present **Research Affiliate**, Harvard T.H. Chan School of Public Health, Boston, MA, USA

Aug 2023–present **Research Affiliate**, Stanley Center, Broad Institute of Harvard and MIT, Cambridge, MA, USA

EDUCATION

Oct 2018–Sep 2023 **D.Phil., Artificial Intelligence for Biomedical Sciences**
University of Oxford, Balliol College, Oxford, UK
National Institutes of Health (NIH) Oxford-Cambridge Scholar
Doctoral Advisors: J. Alison Noble (Oxford/Dept. of Engineering Sciences)
Jeffrey S. Diamond (NIH/Neurological Diseases and Stroke)

Aug 2014–May 2018 **B.Sc., Electrical Engineering**, *Summa Cum Laude*
University of Florida (UF), Gainesville, FL, USA
John V. Lombardi Scholar; GPA: 3.97/4.00
Honors Advisor: Jose C. Principe (UF/Dept. of Electrical Engineering)

POSTDOCTORAL AND ADVANCED TRAINING

Jan 2025–Oct 2026 **Research Fellow in Precision and Genomic Medicine**
Massachusetts General Hospital (MGH) & Harvard Medical School, Boston, MA, USA
NIH/NHGRI T32 Fellowship in Precision and Genomic Medicine
Mentors: Jordan W. Smoller (MGH/Dept. of Psychiatry)
Alkes L. Price (Harvard/T.H. Chan School of Public Health)

Sep 2023–Oct 2026 **Postdoctoral Researcher in Precision Psychiatry**
Massachusetts General Hospital & Harvard Medical School, Boston, MA, USA
Mentors: Jordan W. Smoller (MGH/Dept. of Psychiatry)

RESEARCH CONTRIBUTIONS

AI-Based Psychiatric Phenotyping and Clinical Prediction

Developing and validating AI methods for clinically meaningful measurement of psychiatric outcomes from longitudinal health data.

- Developed and validated an LLM-based pipeline to estimate depression severity from outpatient psychiatric notes, achieving high agreement with clinician assessments and predictive performance comparable to PHQ-9 scores.
- Investigating the integration of longitudinal clinical measures and genetic predictors for psychiatric risk stratification (proposed in the pending K01 application).

Statistical Genetics and Contextualized Risk Prediction

Investigating how environmental and sociodemographic contexts affect the interpretation and portability of genetic risk estimates.

- Led a study across seven diseases showing substantial contextual variation in relative genetic risk estimates but more modest differences in absolute risk.
- Contributed to research on associations of psychiatric genetic liability with metabolic and inflammatory profiles, independent of diagnosis and medication exposure.

Causal Inference and Psychiatric Epidemiology

Applying causal inference and observational study designs to evaluate psychiatric risk factors while addressing confounding.

- Led an international study of COVID-19 containment policies and mental health, finding that lifting restrictions produced only short-term reductions in psychological distress.

Machine Learning Methods and Biomedical Imaging

Developing machine learning methods for noisy biomedical images, limited labeled data, and multimodal reasoning.

- Developed a generative adversarial "synthetic microscope" to create realistic 3D fluorescence microscopy images from neuronal reconstructions, providing labeled synthetic data for image analysis.
- Developed convolutional neural networks for neuron segmentation in low signal-to-noise 3D fluorescence microscopy images.
- Developed deep learning approaches combining plant images and genomic data to predict genotypes from observable phenotypes.
- Developed frameworks to evaluate visual question answering and attention-based multimodal reasoning, and recurrent kernel-based methods for image classification with limited labeled samples.

RESEARCH FUNDING AND FELLOWSHIPS

Pending Grant Applications

2026 **NIH/NIMH K01 Career Development Award**
Leveraging Clinical Notes and Artificial Intelligence for Prognostic Risk Stratification in Major Depressive Disorder
Principal Investigator: Mihael Cudic.
Mentors: Jordan W. Smoller, Ronald C. Kessler, Sabine Wilhelm, Tian Ge, and Alisa Busch.
Proposed development and validation of LLM-based longitudinal depression severity measures, cross-ancestry polygenic scores for the longitudinal course of depression, and interpretable models integrating clinical and genomic data to predict adverse psychiatric outcomes.

Completed Research Fellowships and Support

Jan 2025–Nov 2026 **NIH/NHGRI T32 Fellowship in Precision and Genomic Medicine**
Massachusetts General Hospital and Harvard Medical School
Fellow: Mihael Cudic.
Mentors: Jordan W. Smoller and Alkes L. Price.
Research focused on improving the portability of polygenic risk estimates across diverse populations and sociodemographic contexts.

Aug 2018–May 2023 **NIH Oxford–Cambridge Scholars Program**
University of Oxford and National Institutes of Health
Fellow: Mihael Cudic.
Advisers: J. Alison Noble and Jeffrey S. Diamond.
Doctoral research developing AI methods for fluorescence microscopy analysis of retinal neurons.

PEER-REVIEWED JOURNAL ARTICLES

Cudic M, Tubbs JD, Ge T, Smoller JW. Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk. *American Journal of Human Genetics*. 2026;113(5):966–977. doi:10.1016/j.ajhg.2026.03.013.

Cudic M, Diamond JS, Noble JA. Unpaired mesh-to-image translation for 3D fluorescent microscopy images of neurons. *Medical Image Analysis*. 2023;86:102768. doi:10.1016/j.media.2023.102768.

Cudic M, Burt R, Santana E, Principe JC. A flexible testing environment for visual question answering with performance evaluation. *Neurocomputing*. 2018. Publication link.

Rodriguez MC, Yongye AB, **Cudic M**, Martinez Mayorga K, Liu E, Mueller BM, Ainsley J, Karabencheva-Christova T, Christov CZ, Cudic M, Cudic P. Targeting cancer-specific glycans by cyclic peptide lectinomimics. *Amino Acids*. 2017. Publication link.

PEER-REVIEWED CONFERENCE PAPERS AND PROCEEDINGS

Cudic M, Principe JC. Using a Recurrent Kernel Learning Machine for Small-Sample Image Classification. In: *International Joint Conference on Neural Networks*. 2019:1–6. Publication link.

Cudic M, Baweja HS, Parhar T, Nuske S. Prediction of sorghum bicolor genotype from in-situ images using autoencoder-identified SNPs. In: *IEEE International Conference on Machine Learning and Applications*. 2018:23–31. Publication link.

Burt R, **Cudic M**, Principe JC. Fusing Attention with Visual Question Answering. In: *International Joint Conference on Neural Networks*. 2017:949–953. Publication link.

BOOK CHAPTERS

Cudic M, Diamond JS, Noble JA. Neuron Segmentation from Noisy Fluorescence Microscopy Images Using Deep Learning with Local and Global Scale Fusion. In: *Fluorescence Imaging of the Brain. Neuromethods*. 2024;209:59–81. doi:10.1007/978-1-0716-4011-1_3.

PREPRINTS

Watts D, Khadse PN, Ebrahimi O, Tubbs J, Lian J, Dall'Aglio L, Fatori D, Zhou Y, Zuccolo P, **Cudic M**, De La Hoz Gomez JF, Lee YH, Manfro GG, Bauermeister S, Brunoni A, Choi KW, Kennedy CJ, Smoller JW. Joint Effects of Early Pandemic Containment Policies on Anxiety in the United States. *medRxiv*. 2026. Publication link.

De la Hoz JF, Lee YH, Tubbs JD, Meyerson W, **Cudic M**, Watts D, Feng YCA, Chen Y, Lasky-Su J, Ge T, Smoller JW. Genetic Liability to Psychiatric Disorders is Associated with Metabolic Dysregulation Independent of Diagnosis and Treatment. *medRxiv*. 2026. Publication link.

Cudic M[*], Meyerson W[*], Wang B, Yin Q, Khadse PN, Burke T, Kennedy CJ, Smoller JW. Multi-Criteria Validation of LLM-Inferred Depression Severity from Outpatient Psychiatry Notes. *medRxiv*. 2026. Publication link.

Cudic M[*], de la Hoz JF[*], Dall'Aglio L, Tubbs JD, Ebrahimi OV, Madsen E, Fatori D, Zuccolo PF, Lian J, Kabir D, Zhou Y, Watts D, Choi KW, Manfro GG, Sweeney E, Lin YF, Fancourt D, COVID-19 Global Mental Health Consortium (CGMHC), Patel V, Kessler RC, Bauermeister S, Brunoni AR, Lee YH, Smoller JW. Would Lifting versus Maintaining Covid-19 Containment Policies Have Reduced Psychological Distress in the United States?. *medRxiv*. 2026. Publication link.

Lian J, Zuccolo PF, Ebrahimi OV, Fatori D, Adlakha A, De La Hoz JF, Lee YH, Carneiro A, Benseñor IM, Lotufo PA, Goulart AC, Tubbs JD, Watts D, Zhou Y, Dall'Aglio L, **Cudic M**, Kuenzi M, CGMHC Consortium, Kessler RC, Patel V, Brunoni A, Smoller JW, Bauermeister S. A robust framework for harmonising health measures across international cohorts: Evidence from the COVID-19 pandemic. *medRxiv*. 2025. Publication link.

[*]Equal first-author contribution.

THESES AND DISSERTATIONS

Cudic M. Fluorescence microscopy image analysis of retinal neurons using deep learning. D.Phil., University of Oxford, 2023.

HONORS, AWARDS AND DISTINCTIONS

Postgraduate

2024 — Early Career Investigator Program Award, World Congress of Psychiatric Genetics
Competitive early-career research award (1 of 27 selected; international).

2024 — Best Poster Presentation, World Congress of Psychiatric Genetics
Research presentation award (1 of 3 selected; conference-wide).

Graduate

2018–2023 — NIH Oxford–Cambridge Scholars Program
Fully funded doctoral research fellowship (1 of 11 selected; U.S.-wide).

2018 — National Science Foundation Graduate Research Fellowship Awardee
Graduate STEM research fellowship (1 of ∼2,000 selected; U.S.-wide).

Undergraduate

2018 — Outstanding Four-Year Scholar, University of Florida
Premier undergraduate academic and research distinction (1 of 32 selected; university-wide).

2018 — Electric E Award, University of Florida
Premier undergraduate electrical engineering honor (1 of 7 selected; department-wide).

2017 — Robotics Institute Summer Scholar, Carnegie Mellon University
Undergraduate summer research program focused on robotics and artificial intelligence (1 of 35 selected; international).

2017 Barry Goldwater Scholarship
Undergraduate STEM research scholarship (1 of 240 selected; U.S.-wide).

2017 University Scholar Award, University of Florida
Undergraduate research award recognizing outstanding scholarly achievement.

2014–2018 Dean's List (all 8 semesters), University of Florida
Undergraduate academic distinction (GPA $\geq$ 3.20/4.00).

2014–2018 John V. Lombardi Scholars Program, University of Florida
Four-year undergraduate merit-based scholarship (1 of 8 selected; university-wide).

INVITED TALKS, SEMINARS AND LECTURES

Local and Institutional Seminars

2026 *Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk*; Center for Genomic Medicine, Massachusetts General Hospital, Boston, MA, USA
Oral presentation.

2026 *Large Language Models for Longitudinal Depression Severity Phenotyping*; Center for Digital Mental Health, Massachusetts General Hospital, Boston, MA, USA
Oral presentation.

2022 *Generative Adversarial Networks for Microscopy Images*; Medical Image Analysis Seminar Series, University of Oxford, Oxford, UK
Oral presentation.

2021 *Using Machine Learning to Study Starburst Amacrine Cells*; NIH Sensory Neuroscience Seminar Series, Washington, DC, USA
Oral presentation.

2016 *Pre-Harvest Model for Aspergillus flavus in India*; Department of Plant Sciences, University of Cambridge, Cambridge, UK
Oral presentation.

National

2018 *Prediction of Sorghum bicolor Genotype from In-situ Images Using Autoencoder-identified SNPs*; 17th IEEE International Conference on Machine Learning and Applications, Orlando, FL, USA
Oral presentation.

International

2026 *Large Language Models for Longitudinal Depression Severity Phenotyping*; EDGE Lab, University of Oxford, UK
Oral presentation.

2026 *Large Language Models for Longitudinal Depression Severity Phenotyping*; Statistical Genetics Unit, King's College London, UK
Oral presentation.

2026 *Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk*; Statistical Genetics Unit, King's College London, UK
Virtual presentation.

2026 *Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk*; Institute of Biological Psychiatry, Copenhagen University Hospital, Copenhagen, Denmark
Virtual presentation.

2025 *Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk*; World Congress of Psychiatric Genetics, Cancún, Mexico
Oral symposium presentation.

2024 *Estimating the Causal Effect of Individual Social Distancing Policies on Mental Health Outcomes: A Target Trial Approach*; COVIDMENT Annual Meeting, Stockholm, Sweden
Virtual presentation.

2024 *Estimating the Causal Effects of Social Distancing Policies on Mental Health Outcomes: A Target Trial Approach Across 50 US States*; COVID Global Mental Health Consortium Meeting, Boston, MA, USA
Virtual presentation.

2022 *Understanding Artificial Intelligence from the Lens of Wittgenstein*; Balliol College Reading Course, Les Houches, France
Oral presentation.

2019 *Automatic Segmentation of Overlapping Neurons Using Convolutional Neural Networks*; Max Planck / HHMI Connectomics Meeting, Berlin, Germany
Oral presentation.

CONFERENCE ABSTRACTS AND POSTER PRESENTATIONS

2026 *Prevalence Differences in Major Depressive Disorder Explain Apparent Heterogeneity in Polygenic Score Effect Sizes*; World Congress of Psychiatric Genetics, Glasgow, Scotland, UK

2025 *Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk*; American Society for Human Genetics, Cancún, Mexico

2024 *Polygenic Scores across Environmental Contexts: How Intersectional Factors Affect Predictive Accuracy for Psychiatric Conditions*; World Congress of Psychiatric Genetics, Singapore
Best poster winner

2018 *Using a Recurrent Kernel Learning Machine for Low-Sample Face Detection*; University of Florida Undergraduate Research Symposium, Gainesville, FL, USA

2017 *Using a Locally Connected Autoencoder to Identify Candidate Alleles*; Carnegie Mellon University Robotics Institute Summer Scholars Conference, Pittsburgh, PA, USA

2017 *Fusing Attention with Visual Question Answering*; 30th International Joint Conference on Neural Networks, Anchorage, AK, USA

2015 *Molecular Modeling Studies of Peptide-Based Lectinomimics*; 23rd American and 6th International Peptide Symposium, Waikoloa Village, HI, USA

TEACHING EXPERIENCE

Course Instruction

Jan–May 2024 Course Director and Instructor, *AI Applications in Psychiatry*; University of Florida Honors Program
Designed curriculum spanning machine learning, psychiatric research, genetics, a case-based international clinical discussion with a psychiatrist, and an applied capstone using public mental health data. Average student rating: 4.8/5.0; course showcased and recognized by the Honors Program.

Oct–Dec 2022 Teaching Assistant, *Engineering Computation Laboratory*; University of Oxford
Guided students in implementing and debugging computational and AI algorithms.

Aug–Dec 2017 Teaching Assistant, *Elements of Machine Intelligence*; University of Florida
Prepared teaching materials, graded coursework and examinations, and held office hours for graduate-level machine learning.

Jan–May 2016 Teaching Assistant, *Circuits*; University of Florida
Independently developed and assessed the laboratory component of an undergraduate electrical engineering course.

Research Mentorship and Supervision

Mar 2026–present **Ashley Danskee**, Clinical Research Coordinator
Provided training in statistical analysis and AI methods for psychiatric research, and guidance on clinical psychology graduate applications.

Jun–Oct 2024 **James Hu**, Undergraduate Researcher, University of Florida
Supervised methodological development for gene–environment interaction analyses; subsequently participated in the Broad Institute summer research program and pursued graduate studies in computational biology.

EDITORIAL AND PEER-REVIEW SERVICE

Ad hoc Reviewer

2026–present BMC Psychiatry
Peer reviewer

2017 IEEE Symposium Series on Computational Intelligence
Peer reviewer

PROFESSIONAL SOCIETY MEMBERSHIPS

2023–present Member, International Society of Psychiatric Genetics

2024–present Member, American Society for Human Genetics

PUBLIC EDUCATION AND OUTREACH

Public Science Communication

Ongoing — Public science communication Maintains `https://www.mihaelcudic.com/` with accessible explanations of AI, genetics, and psychiatry research; engages in panels on interdisciplinary research, graduate education, and academic careers.

Public Lectures and Educational Media

2025 — *From Code to Clinic: A History of AI in Medicine* Invited educational lecture for the Broad Institute's Medical and Population Genetics (MPG) Primer series; over 1,000 views on YouTube.

MEDIA COVERAGE AND RESEARCH FEATURES

Personal Features

2017 "UF Student Earns Goldwater Scholarship," University of Florida News
Institutional news feature highlighting selection as a Barry Goldwater Scholar.

OTHER EMPLOYMENT AND RESEARCH EXPERIENCE

Jan 2015–May 2018 **Undergraduate Researcher and University Scholar**, Computational NeuroEngineering Laboratory, University of Florida, Gainesville, FL, USA
Developed custom machine learning algorithms for image classification under limited training data conditions.

Jun–Aug 2017 **Machine Learning Intern and Summer Scholar**, Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA
Developed deep learning models to predict plant genotypes from in situ imaging data.

Jun–Aug 2016 **Visiting Researcher**, Department of Plant Sciences, University of Cambridge, Cambridge, UK
Developed stochastic compartmental models to study and predict infectious disease transmission in plants.

Jun–Aug 2014 **Statistics Intern**, United States Department of Agriculture, Fort Pierce, FL, USA
Developed software to analyze citrus greening disease risk across large agricultural fields.

Jun–Aug 2013 **Computational Chemistry Intern and Student Scholar**, Torrey Pines Institute for Molecular Studies, Port St. Lucie, FL, USA
Performed conformational analysis of cyclic peptide analogues using computational methods.

SELECTED PROFESSIONAL DEVELOPMENT

2025 Genetics Training Program, Harvard Medical School, Boston, MA, USA
Advanced training in clinical genetics, molecular genetics, and genomic medicine (in person).

2025 Computational Genomics Summer Institute, University of California, Los Angeles, CA, USA
Advanced training in computational genomics, population genetics, and statistical methods (in person).

2019 International Computer Vision Summer School, University of Catania, Catania, Italy
Advanced training in computer vision, image analysis, and machine learning (in person).

RESEARCH INTERESTS

Artificial Intelligence • Computational Psychiatry • Digital Mental Health • Longitudinal Psychiatric Phenotyping • Transdiagnostic Psychiatry • Psychiatric Genetics • Genetic Risk Prediction • Statistical Genetics • Mental Health Epidemiology • Causal Inference • Cognitive Neuroscience • Neuroimaging

TECHNICAL SKILLS AND METHODS

Artificial Intelligence	Large Language Models • Natural Language Processing • Generative AI • Deep Learning • Computer Vision • Medical Image Analysis • Image Segmentation • Autoencoders
Statistical Genetics	Genome-Wide Association Studies • Polygenic Scores • Heritability Estimation • Genetic Risk Prediction • Gene–Environment Interactions
Psychiatry	Computational Psychiatry • Psychometrics • Electronic Health Records • Clinical Phenotyping • Longitudinal Data Analysis
Epidemiology	Causal Inference • Target Trial Emulation • Synthetic Control Methods • Difference-in-Differences • Stochastic Compartmental Models
Experimental Methods	Fluorescent Labeling • Transgenic Mouse Models • Fluorescence Microscopy • Tissue Dissection

REFERENCES

Postdoctoral Mentor
Jordan W. Smoller, MD, ScD
Professor of Psychiatry, Harvard Medical School
Director, Center for Precision Psychiatry, Massachusetts General Hospital
jsmoller@mgh.harvard.edu

Fellowship Mentor
Alkes L. Price, PhD
Professor of Statistical Genetics, Harvard T.H. Chan School of Public Health
Department of Epidemiology
aprice@hsph.harvard.edu

Doctoral Adviser
J. Alison Noble, DPhil
Technikos Professor of Biomedical Engineering
Department of Engineering Science, University of Oxford
alison.noble@eng.ox.ac.uk

Doctoral Adviser
Jeffrey S. Diamond, PhD
Scientific Director and Senior Investigator
National Institute of Neurological Disorders and Stroke, NIH
jeffrey.diamond@nih.gov

EXTENDED RESEARCH PROFILE

Mihael Cudic is a computational biomedical researcher working at the intersection of artificial intelligence, statistical genetics, epidemiology, neuroscience, and psychiatry. He earned a B.S. in Electrical Engineering, summa cum laude, from the University of Florida and completed interdisciplinary doctoral training at the University of Oxford and the U.S. National Institutes of Health through the NIH Oxford–Cambridge Scholars Program. His doctoral research introduced machine-learning approaches to analyzing noisy three-dimensional fluorescence microscopy of retinal neurons.

Following his D.Phil., he joined the Center for Precision Psychiatry at Massachusetts General Hospital, where he investigates the use of statistical and AI methods for psychiatric clinical measurement and risk prediction. His projects include causal analyses of COVID-19 containment policies and mental health, use of large language models to assess depression severity from clinical notes, and the contextual portability of polygenic scores. His first-author 2026 *American Journal of Human Genetics* article examined variation in relative and absolute genetic risk across intersecting contexts.

His teaching has spanned engineering, machine learning, and biomedical science. He independently developed the University of Florida Honors course *AI Applications in Psychiatry* and has supervised students entering computational biomedical research. His long-term research goal is to develop interpretable, rigorously validated AI-enabled tools integrating clinical and genomic data to improve psychiatric risk stratification and support clinically meaningful decisions.